

Resource-Normalized Scaling Laws for Byzantine Consensus: Chance-Constrained Validator-Set Sizing from Commodity-CI Measurements

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Problem. Sizing a permissioned BFT validator set for national-scale e-governance must reconcile degrading consensus throughput with improving fault-detection power as n grows; without reporting the concurrency path, published single-factor exponents are difficult to interpret as intrinsic protocol properties. **Contributions.** An engineering methodology for resource-normalized capacity planning: *method* — resource-normalized measurement isolating consensus cost from host contention; *analysis* — identifiability relation $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ along a concurrency path; *engineering* — chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$; *empirics* — coverage-checked prediction intervals on public CI. **Validation.** A four-core commodity-CI campaign covers $n \leq 16$, concurrency up to 64, emulated RTT, and omission/duplication fault injection; fitted coefficients are deployment-specific.

Keywords: *Byzantine fault tolerance, consensus, scaling law, capacity planning, chance-constrained optimization, e-governance*

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1. Introduction

Permissioned Byzantine Fault Tolerant (BFT) consensus underpins e-governance systems — voting, registries, digital identity — and raises a concrete procurement question: *how many validators should the network have?*

The validator count n pulls in opposite directions. Throughput argues for small n : every validator executes every transaction, and PBFT-family protocols exchange $O(n^2)$ messages per block. Detection reliability argues for large n . This paper develops a capacity-planning methodology that combines the two under resource-normalized measurement.

Published single-factor fits $\text{TPS}(n) \propto n^\alpha$ disagree in magnitude and sometimes in sign. Our earlier studies, extrapolated to sixty validators, differ by roughly an order of magnitude. Such disagreement is often read as protocol-specific behaviour. Within the investigated setting we show that it can arise from measurement design alone: concurrency is rarely held fixed while n varies, so the fitted scaling exponent mixes load-generator growth with consensus overhead. Without reporting the concurrency path, published exponents are difficult to interpret as intrinsic protocol properties. Packing many validators onto one host adds a second confound — CPU contention — that resource-normalized measurement is designed to isolate.

Contributions. Methodologically, we introduce a resource-normalized measurement (RNM) protocol that fixes CPU quota q per validator and reports $\widetilde{\text{TPS}} = \text{TPS}/q$, separating consensus cost from host contention. Analytically, under the unsaturated bi-factor model, along a concurrency path of slope λ the single-factor OLS fit satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Proposition 1); the relation is a consequence of the model assumptions, not an empirical universal. From a planning perspective, we formulate chance-constrained sizing whose feasible set is $[n_{\min}, n_{\max}]$ (Proposition 2). Empirically, we check prediction-interval coverage and exercise the workflow in a public Besu/QBFT CI campaign (Sections 5–6), including a path check of consistency with Proposition 1 within that configuration. Scope: throughput and detection under the investigated deployment; absolute TPS are environment-specific; cross-client replication is out of scope. Reported coefficients are deployment-specific; the methodology is intended to transfer under matching measurement conditions.

2. Related work

BFT protocols. Classical PBFT [1] tolerates $f < n/3$ at $O(n^2)$ messages [2, 3]. Later designs — HotStuff [4], Tendermint [5], Algorand [6], HoneyBadgerBFT [7] — trade latency and randomization; Vukolić [8] argues that no protocol dominates. These lines of work fix the validator set; they do not size it under joint throughput and detection requirements.

Benchmarks. Fabric [9, 10], BLOCKBENCH [11] and BFT-SMaRt [12] report absolute throughput at chosen configurations. They do not separate concurrency from validator count, nor state measurement conditions under which fitted exponents would be expected to transfer.

Scalability laws and tools. Amdahl [13], Gustafson [14] and Gunther’s USL [15] fold concurrency and system size into one load variable — the confound that RNM is intended to remove. Concentration bounds [16, 18], Kish’s design effect [17], delta-method intervals [20], the residual bootstrap [19] and chance-constrained programming [21, 22] are standard instruments; the present framework couples throughput and detection through opposing monotonicities in n .

Prior work. Remote-voting analyses [23, 24, 25] place endpoint and coercion threats out of scope. Our earlier calibration [27] and switching study [28] left open an identifiable consensus exponent and sizing with measured detector rates.

3. Resource-normalized measurement and the bi-factor law

Capacity planning for BFT benefits from an exponent that reflects consensus cost rather than host contention or load-generator growth. We introduce an isolation protocol, a bi-factor law, and an identifiability result under stated assumptions.

3.1. Resource-normalized measurement as an isolation tool

If n validators share a host of Q cores, each receives Q/n cores, so measured throughput mixes algorithmic cost with a shrinking compute budget. RNM is designed to isolate those effects under controlled quotas.

Definition 1 (Resource-normalized measurement). *A campaign follows the RNM protocol with quota q if every validator container receives the same CPU quota and memory limit independently of n , with aggregate reservation nq below host capacity. Normalized throughput is $\text{TPS} := \text{TPS}/q$.*

Assumption 1 (Separable compute budget). On the operating range of quotas in the campaign, $\text{TPS}(n, c, q) = q \cdot \Phi(n, c) \cdot (1 + r)$ with $|r| \leq Cq$ for a constant C independent of (n, c) there.

Assumption validation. Under Assumption 1 and a power-law factor $\Phi \propto n^{-\beta}$, the OLS slope of $\log \text{TPS}$ on $\log n$ at fixed concurrency is independent of q : the relative remainder is absorbed by the intercept. Section 5 tests this invariance across $q \in \{0.20, 0.25, 0.33\}$. Within the investigated campaign the model is bounded to the reference quota $q = 0.25$; outside that regime the numerical law must be re-fit.

Definition 2 (Bi-factor scaling law). Under the RNM protocol,

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = \frac{c^\gamma}{1 + (c/c^*)^\theta}, \quad (1)$$

with $T_0 > 0$, $\beta > 0$, $\gamma \in (0, 1]$, $\theta > \gamma$, $c^* > 0$. In the unsaturated regime $c \ll c^*$ this is log-linear: $\log \text{TPS} = \log T_0 + \gamma \log c - \beta \log n$.

The form separates concurrency from validator-count scaling, reduces to a power law for $c \ll c^*$, saturates at large c , and contains the classical single-factor law as the case $\gamma = 0$ (or $\lambda = 0$ with $\alpha = -\beta$).

3.2. Identifiability

Proposition 1 is an analytical result under the unsaturated bi-factor model and the stated path and noise assumptions. Its practical value is an *engineering interpretation*: discrepant single-factor BFT exponents can arise from the concurrency schedule rather than from the protocol alone. The campaign in Section 5 checks consistency with the predicted line within the investigated configuration; it does not establish universal validity beyond those assumptions.

Proposition 1 (Identifiability of the consensus exponent). Let observations follow the unsaturated bi-factor law with additive noise of mean zero and finite variance, along a concurrency path $\log c_i = \lambda \log n_i + \mu + \eta_i$ ($\mathbb{E}[\eta_i | n_i] = 0$). If the empirical second moment of $x_i = \log n_i$ converges to a positive limit, the OLS slope $\hat{\alpha}_m$ of $\log \text{TPS}_i$ on x_i alone satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$. In particular $\hat{\alpha} > 0 \iff \lambda > \beta/\gamma$, and $\hat{\alpha} = -\beta \iff \lambda = 0$.

Proof. Write $x_i = \log n_i$, $y_i = \log \text{TPS}_i$. The path reduces the unsaturated law to $y_i = a + \alpha x_i + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i | x_i] = 0$, finite variance, and

$$\alpha = -\beta + \gamma\lambda. \quad (2)$$

Centred OLS gives $\hat{\alpha}_m = \alpha + N_m/D_m$ with $N_m = \sum (x_i - \bar{x})\varepsilon_i$ and $D_m = \sum (x_i - \bar{x})^2$. The second-moment assumption yields $D_m/m \rightarrow \text{Var}^*(x) > 0$; the summands of N_m are uncorrelated, mean zero, and of bounded second moment, so $N_m/m \rightarrow 0$ in probability. Continuous mapping: $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \alpha$. Signs follow from $\gamma > 0$. ■

Thus $\alpha = \underbrace{-\beta}_{\text{protocol}} + \underbrace{\gamma\lambda}_{\text{measurement}}$. Only a factorial design ($\lambda = 0$) recovers the consensus exponent under the stated model. For illustration, $\gamma = 0.8$, $\beta = 0.5$ and $\lambda = 1$ yield $\hat{\alpha} = +0.3$, and extrapolating from $n = 5$ to $n = 60$ differs by a factor ≈ 7 , matching the order of the discrepancy in [27].

Corollary 1 (Non-transferability of single-factor exponents). *Under the unsaturated bi-factor law, laboratories measuring along $\lambda_1 \neq \lambda_2$ obtain exponents differing by $\gamma(\lambda_1 - \lambda_2)$. Without reporting λ , published single-factor exponents should not be treated as transferable protocol properties across measurement designs.*

3.3. Implementation

On a factorial grid ($\lambda = 0$) we estimate $(\log T_0, \gamma, \beta)$ by OLS in log-space and (c^*, θ) by nonlinear least squares on the full c -sweep (Section 5). The calibrated law then feeds prediction intervals and the sizing program of the next section within the range supported by the campaign.

4. Prediction intervals, detection, and the sizing problem

An identifiable exponent is an input to capacity planning. This section assembles a sizing framework: a throughput chance constraint and an optional detection constraint. Any detector yielding measured (p_d, p_f, ρ) plugs in; X5 is one instance.

4.1. Definitions and assumptions

Definition 3 (Fault class and detector). *A subset B of size k is faulty if each member fails a proposal round independently with probability $p_d > p_f$ (omission class; equivocation excluded). The detector flags validator i when its participation over w rounds falls below $\tau \in (p_f, p_d)$; detection succeeds when at least one member of B is flagged.*

Assumption 2 (Exchangeable correlation). Participation indicators in B are exchangeable with pairwise correlation $\rho \in [0, 1)$, so the mean carries design effect $1 + (k - 1)\rho$ in the sense of Kish [17].

4.2. Prediction intervals and detection bound

With regressor $\mathbf{z} = (1, \log c, -\log n)^\top$ and $\hat{\vartheta} = (\log T_0, \gamma, \beta)^\top$, the delta method [20] gives predictive variance $\hat{V} = \mathbf{z}^\top \widehat{\text{Cov}}(\hat{\vartheta}) \mathbf{z} + \hat{\sigma}^2$. An approximate $(1 - \varepsilon_p)$ interval for $\text{TPS}(n, c)$ is then $\exp\{\log \widehat{\text{TPS}} \mp z_{1-\varepsilon_p/2} \hat{V}^{1/2}\}$; the one-sided lower endpoint $\underline{\text{TPS}}_{1-\varepsilon_p}$ uses $z_{1-\varepsilon_p}$. A residual bootstrap [19] supplies empirical quantiles when residual normality is doubtful. Coverage is checked on held-out validator counts (Section 5).

With $k = \lceil \varphi n \rceil$, a design-effect heuristic yields the approximate miss probability

$$p_{\text{miss}}(n, w) \lesssim \exp\left\{-\frac{2wk(p_d - \tau)^2}{1 + (k - 1)\rho}\right\}, \quad (3)$$

which is nonincreasing in n and w . The cut $p_{\text{miss}} \leq \varepsilon_s$ gives

$$n_{\min} = \left\lceil \frac{1 + (k - 1)\rho}{\varphi} \cdot \frac{\log(1/\varepsilon_s)}{2w(p_d - \tau)^2} \right\rceil. \quad (4)$$

This adapts Hoeffding [16] via Kish [17] with $n_{\text{eff}} = kw/(1 + (k - 1)\rho)$. With X5 rates (AUC = 0.506), the campaign defaults to the throughput-bound constraint.

4.3. The sizing program

Problem 1 (Validator-set sizing). Given peak demand D_{peak} , levels $\varepsilon_p, \varepsilon_s$, window w , faulty fraction φ , production scale-up $\kappa > 0$ from RNM units to absolute throughput, and nondecreasing $\text{Cost}(n)$, minimize $\text{Cost}(n)$ subject to $\mathbb{P}[\kappa \cdot \widetilde{\text{TPS}}(n, c^*) \geq D_{\text{peak}}] \geq 1 - \varepsilon_p$ and $p_{\text{miss}}(n, w) \leq \varepsilon_s$, where $c^* = \arg \max g$. When κ is free, take the smallest value that meets the throughput constraint at $n = 4$.

The chance constraint is equivalent, via the prediction interval above, to

$$h(n) := \log(\kappa T_0 g(c^*)/D_{\text{peak}}) - \beta \log n - z_{1-\varepsilon_p} \hat{V}^{1/2}(n, c^*) \geq 0. \quad (5)$$

On the extrapolation range the fitted covariance confirms that $\hat{V}$ is nondecreasing in $\log n$ (Section 5), so with $\beta > 0$ the map h is strictly decreasing there; on the measured grid we evaluate h directly.

Proposition 2 (Feasibility window). *Whenever h is strictly decreasing and the approximate detection cut (4) is used, the feasible set is $\{n : n_{\min} \leq n \leq n_{\max}\}$ with $n_{\min}$ from (4) and $n_{\max} = \lfloor \sup\{n : h(n) \geq 0\} \rfloor$. If nonempty and Cost is nondecreasing then $n^* = n_{\min}$.*

Proof. Detection holds on $\{n \geq n_{\min}\}$ and throughput on $\{n \leq n_{\max}\}$, hence an interval; monotone cost selects the left endpoint. ■

Corollary 2 (Empty feasible set). *If $n_{\min} > n_{\max}$, no validator count satisfies both requirements under the stated constraints: the architecture must change, not merely the provisioning.*

The window is the intersection of two monotone constraints [21, 22]; within the proposed framework it converts calibrated exponents into a sizing rule for the investigated deployment class.

5. Experimental validation

5.1. Setup and design

The campaign runs in a public continuous-integration workflow on a four-core, 16 GB hosted runner. Each job provisions a Hyperledger Besu [26] QBFT network of n validators under the RNM protocol (Definition 1). A closed-loop generator drives c concurrent submitters and records steady-state committed throughput after warm-up. Faults are injected with kernel traffic control (omission and duplication; not equivocation).

X1/X3 calibrate (β, γ) on a factorial grid; **X2** checks the quota regime; **X3b** re-samples along concurrency paths; **X4** varies emulated RTT; **X5** reports detector AUC; **X7/X8** check prediction-interval coverage and sizing ablation. Cross-client replication (**X9**/GoQuorum) is out of scope. In total 94 successful records, ≈ 27.2 core-hours, $n_{\max} = 16$.

Reproducibility. All experiments are reproducible from the public CI workflow and archived artefacts at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

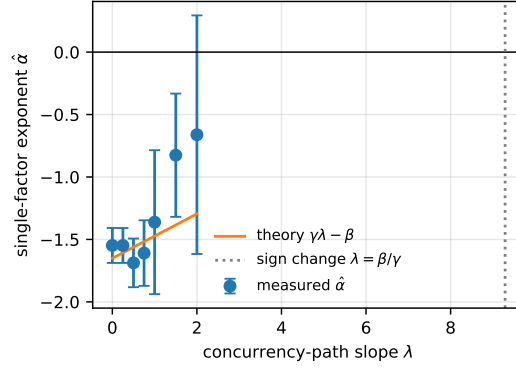


Fig. 1. Measured single-factor exponents track $\gamma\lambda - \beta$ within the investigated configuration, consistent with the analytical prediction of Proposition 1. Sign change predicted at $\lambda = \beta/\gamma = 9.29$

Table 1. Calibrated parameters, invariance, detector, and coverage

Quantity	Estimate	Note
$T_0; \beta; \gamma$	2016.5; 1.651 ± 0.151 ; 0.178 ± 0.070	bi-factor
$c^*, \theta; \hat{\sigma}$	67.9, 2.542; 0.404	saturation / residual
$\hat{\alpha}$ at $\lambda = 0, 1, 2$	-1.548, -1.362, -0.661	X3b
β at $q = 0.20, 0.25, 0.33$	1.550, 1.541, 0.879	X2, $p = 0.002$
β at RTT 0/high	1.527, 2.708	X4
$p_d, p_f, \rho; \text{AUC}$	0.397, 0.376, 0.000; 0.506	X5, $\tau = 0.90$
coverage 90%/95%	90.9 %, 95.5 %	hold-out $n \in \{11, 13, 16\}$

5.2. Results

X1/X3 – bi-factor calibration. Joint OLS on the reference-quota subset yields $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$ ($R^2 = 0.727$; Table 1). Under RNM one expects $\beta > 0$: consensus cost grows with validator count when concurrency is held fixed. The fit confirms that expectation at $q = 0.25$ within the investigated setting.

X3b – path check of Proposition 1. A single-factor fit along $\lambda = 1$ gives $\hat{\alpha} = -1.362$ against the predicted $\gamma\lambda - \beta = -1.473$. Figure 1 shows that measured exponents track $\gamma\lambda - \beta$ across $\lambda \in \{0, 0.5, 1, 2\}$ within the investigated configuration. This is consistency with the analytical prediction under the campaign’s measurement conditions, not a claim of universal validity.

X2 – quota-regime check. The partial F -test for $\text{quota} \times \log n$ interaction rejects pooling across $\{0.20, 0.25, 0.33\}$ ($p = 0.002$): β drops from ≈ 1.541 at $q = 0.25$ to 0.879 at $q = 0.33$. Coefficients are therefore reported at the stable reference quota $q = 0.25$.

X4 – latency. Under elevated emulated RTT, β rises from 1.527 to 2.708. WAN delay steepens the consensus penalty in validator count in this campaign. Sizing rules calibrated on LAN-like RTT are therefore optimistic for geo-distributed deployments unless re-calibrated.

X5 – detector. Omission/duplication detection yields $\text{AUC} = 0.506$ (Figure 2), so sizing defaults to the throughput-bound constraint.

X7/X8 – coverage, USL comparison, and ablation. Hold-out coverage meets the nominal levels in Table 1 for the calibrated range. The fitted covariance confirms that $\hat{V}$ is nondecreasing for $n \geq 16$, so h in (5) is strictly decreasing on the extrapolation range; on the measured grid we evaluate h directly. Out-of-sample log-RMSE for linear / single-factor / USL [15] / bi-factor predictors is 1.329, 0.523, 0.637, 0.538. The bi-factor improvement over USL is consistent with decoupling concurrency from replica count under RNM. Relative sizing costs: 4.00, 1.50, 1.50, 1.00.

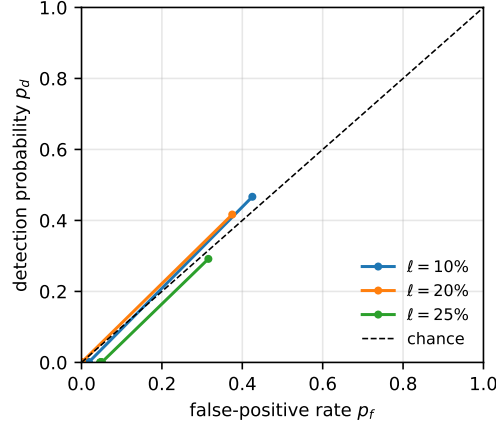


Fig. 2. Detector ROC (AUC = 0.506); throughput binds the window

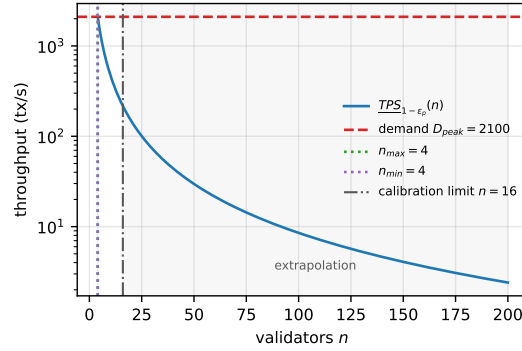


Fig. 3. Scenario A: at minimal admissible $\kappa = 13.00$, the feasibility window is $[n_{\text{min}}, n_{\text{max}}] = [4, 4]$. The dash-dotted line marks the calibration limit $n = 16$; the shaded region to its right is model-based extrapolation

6. Application, implications, and limitations

6.1. Application and what-if analysis

We instantiate Problem 1 on $\approx 3 \times 10^7$ ballots over twelve hours ($D_{\text{peak}} \approx 2.1 \times 10^3$ TPS), requiring detection of a coordinated omission fault of fraction $\varphi = 0.2$ within one epoch at level $1 - \varepsilon_s$. Absolute demand enters via scale-up κ from RNM units. With free κ , Problem 1 takes

$$\kappa^* = \frac{D_{\text{peak}}}{\widetilde{TPS}_{1-\varepsilon_p}(4, c^*)}, \quad (6)$$

the minimal scale at which the lower PI bound at $n = 4$ meets D_{peak} .

Scenario A (baseline). Substituting the calibrated coefficients ($\beta = 1.651$, $\gamma = 0.178$) at $c^* = 32$ yields $\kappa = 13.00$ and feasibility window $[4, 4]$ with $n^* = 4$ (Proposition 2; Figure 3). Under this demand and hardware class the window collapses to a single admissible count: the network is throughput-bound at the smallest BFT set admitted by the formulation.

Scenario B (doubled peak demand, fixed hardware class). Hold $\kappa = 13.00$ and replace D_{peak} by $2D_{\text{peak}}$. Then h in (5) falls by $\log 2$ everywhere; in particular $h(4) < 0$, so the feasible set is empty (Corollary 2). Within the calibrated law, doubling peak load on the same hardware class is an architectural impossibility: adding validators cannot restore feasibility, because throughput falls with n .

Scenario C (WAN deployment, fixed hardware class). Retain $\kappa = 13.00$ and D_{peak} , but substitute the high-RTT exponent $\beta = 2.708$ (X4) for $\beta = 1.527$. The steeper penalty yields $h(4) < 0$ and an empty window: a hardware class that is marginally admissible under LAN β becomes infeasible under WAN β without raising κ .

The workflow applies equally to registry and identity sizing under analogous demand assumptions; binding elections on a blockchain remain outside the recommendation [23, 24, 25].

6.2. What this implies

A fitted (β, γ) transfers when concurrency path, quota regime, and RTT class match the calibration. Our results suggest that future benchmarks should report λ (or $\lambda = 0$), per-validator quota, and RTT class (Corollary 1).

6.3. Limitations

Fault class. Encrypted transport limits X5 to omission/duplication (AUC 0.506; throughput-bound window).

Infrastructure and range. Experiments use commodity four-core CI with $n \leq 16$. Beyond that range, extrapolation is model-based and is quantified by prediction intervals.

Coefficients versus methodology. Numerical (β, γ) are Besu/QBFT-specific. Proposition 1 is analytical under model assumptions; the campaign shows consistency within the investigated configuration. The RNM protocol and chance-constrained workflow are intended to generalize under matching measurement conditions; production scale-up uses κ in (6).

7. Conclusions

What is established. Under the unsaturated bi-factor model and stated path assumptions, the single-factor OLS exponent satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Proposition 1). Opposing monotone constraints yield a closed integer feasibility window (Proposition 2). The contribution is a reproducible engineering methodology for resource-normalized capacity planning: isolate a consensus exponent under RNM, then size the validator set under calibrated chance constraints.

What the experiments indicate. Under RNM on four-core commodity CI: $\beta = 1.651 \pm 0.151$, $\gamma = 0.178 \pm 0.070$; path re-sampling is consistent with $\gamma\lambda - \beta$ within the investigated configuration; coefficients at $q = 0.25$; coverage meets nominal levels within the calibrated range; AUC = 0.506 defaults the window to throughput. WAN RTT raises β from 1.527 to 2.708; doubled demand or WAN β at fixed κ empties the window. Numerical results are deployment-specific.

What remains open. Equivocation-capable instrumentation; multi-client numerical checks; latency or finality constraints inside the window; online recalibration of β .

Conflict of interest

The authors declare no conflict of interest.

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Data availability

Workflow definition, generators, fault-injection and analysis code, and raw artefacts are available at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

Use of artificial intelligence

AI-assisted tools were used for language editing and typesetting. All scientific results, proofs, design and analysis were produced by the authors.

Contributions of authors

Peniak B. O.: conceptualization, methodology, formal analysis, software, investigation, data curation, writing — original draft, visualization. **Liubinskyj B. B.:** supervision, validation, writing — review and editing.

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РЕСУРСНО-НОРМОВАНІ ЗАКОНИ МАСШТАБУВАННЯ BFT-КОНСЕНСУСУ: ІМОВІРНІСНО-ОБМЕЖЕНЕ ВИЗНАЧЕННЯ РОЗМІРУ МНОЖИНИ ВАЛІДАТОРІВ ЗА ВИМІРЮВАННЯМИ НА ТИПОВІЙ СІ-ІНФРАСТРУКТУРІ

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Методика ресурсно-нормованого планування потужності для визначення розміру множини валідаторів BFT-мережі в електронному урядуванні: пропускна здатність спадає зі зростанням n , а надійність виявлення відмов зростає з n . Внески: метод — ресурсно-нормовані вимірювання; аналіз — співвідношення ідентифікованості ($\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$); інженерія — задача з імовірнісними обмеженнями та вікном $[n_{\min}, n_{\max}]$; емпірика — прогнознi інтервали з перевіркою покриття у публічному СІ. Коефіцієнти специфічні для дослідженого розгортання.

Ключові слова: візантійська відмовостійкість, консенсус, закон масштабування, планування потужності, оптимізація з імовірнісними обмеженнями, електронне урядування

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